

AI-Driven QoS Prediction in SD-WAN

Networks



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DECLARATION I, Radhika(Rishiv) Shitlani, hereby declare that this thesis, titled “AI-Driven QoS Prediction in SD-WAN Networks”, and the work presented in it are entirely my own except where explicitly stated otherwise in the text, and that this work has not been previously submitted, in part or whole, to any university or institution for any degree, diploma, or other qualification.

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Abstract

Software-Defined Wide Area Networks (SD-WAN) are increasingly used by enterprises to connect distributed sites over multiple link types. Ensuring consistent Quality of Service (QoS) in these environments is challenging because traffic patterns fluctuate dynamically and existing SD-WAN controllers rely on static, reactive policies that act only after degradation has occurred.

This thesis investigates whether machine learning models can predict QoS outcomes in SD-WAN-like network environments using publicly available datasets, enabling a proactive approach to network management. Three research questions guide the work: what AI and ML techniques are suitable for QoS prediction; what are their accuracy, scalability, and real-world limitations; and how predictions can be integrated into production SD-WAN systems.

A literature review of recent research identifies supervised regression, deep learning, graph neural networks, and reinforcement learning as the main techniques applied to QoS prediction and SD-WAN traffic engineering. The primary dataset used for experimentation is the Zenodo 5G campus network dataset (record 13754300), which contains measured throughput, jitter, packet loss, and one-way delay across two 5G testbeds under varied radio configurations.

Baseline experiments on this dataset show that Linear Regression achieves an R^2 score of 0.895 and a Mean Absolute Error of 10.55

Mbps when predicting achieved throughput, significantly outperforming a naive mean-prediction baseline ($R^2 = -0.007$). Inference time of 0.032 milliseconds per prediction confirms suitability for near-real-time deployment.

Subsequent experiments will extend the evaluation to Random Forest, Gradient Boosting, Support Vector Regression, and feedforward neural networks. Results, limitations, and directions for future deep-learning-based approaches are discussed.

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Chapter 1

Introduction

In today's digital world, businesses rely heavily on wide area networks (WANs) to connect their offices, data centers, and cloud services. Traditional WANs, which often use expensive and rigid MPLS links, are increasingly being replaced by Software-Defined Wide Area Networks (SD-WAN). SD-WAN offers more flexibility, cost savings, and centralized control. One of the biggest responsibilities of any WAN including SD-WAN is to ensure smooth and reliable network performance. This is referred to as Quality of Service (QoS), which includes important metrics like delay (latency), jitter, packet loss, and available bandwidth.

In networks, traffic patterns keep fluctuating since different applications have different needs, and the same network might suddenly be overloaded. For example, video conferencing needs minimal or zero packet loss along with low latency, while file downloads just need higher bandwidth. If this is not taken care of, user experience is impacted and organizations may face penalties for violating SLAs (Service Level Agreements). Therefore, maintaining good QoS is not only critical but challenging.

Currently, most SD-WAN solutions use static policies to configure QoS. This means that network policies are either set manually in advance or adjusted after

a problem is detected or reported. The problem here is: by the time a QoS issue is noticed, it might already be too late and user experience is already impacted. This delay can affect real-time applications like voice or video calls, which need instant action. There's a clear need for a better, smarter solution.

This is where Artificial Intelligence (AI) and Machine Learning (ML) come in and can help to resolve the problem. These technologies can learn from historical data and detect patterns in the network traffic. If trained correctly, ML models can predict QoS problems before they happen. For example, if the system learns that high video traffic around 2 PM usually leads to congestion, it can automatically adjust bandwidth or reroute traffic ahead of time. This kind of prediction allows network operators to prevent problems instead of just reacting to them.

Scope and Context

This report explores how AI and ML techniques are being used to predict QoS in SD-WAN networks. Instead of automating policy enforcement (which was part of our earlier plan), the focus here is only on prediction specifically, predicting QoS metrics like latency or bandwidth availability using real traffic data. I tried look at several research papers that have used ML models (like decision trees, neural networks, or reinforcement learning) to make these predictions, and examine how accurate and practical those solutions are.

Problem Statement

SD-WAN makes it easier to manage WANs, but maintaining QoS is still a challenge, especially during unexpected traffic surges or link failures. Most QoS-related decisions today are made either too late or based on static rules. Predicting QoS metrics in advance using AI could solve this problem but it's not clear which methods work best, how accurate they are, or how easy they are to

implement in real networks.

Research Questions

This report aims to answer the following questions:

- **RQ1:** What types of AI or ML techniques are currently or can be used for predicting QoS in SD-WAN?
- **RQ2:** What are the benefits and limitations of these approaches in terms of prediction accuracy, scalability, and real-world use?
- **RQ3:** How can QoS prediction be effectively integrated into SD-WAN production environments to improve network performance?

Methodology

To explore these questions, I conducted a literature review using academic databases like IEEE Xplore, Scopus, and Google Scholar. I searched for terms such as “QoS prediction,” “SD-WAN,” “machine learning in networking,” and “AI-based traffic forecasting.” Papers published between 2020 and 2024 were prioritized, and older foundational works were included when relevant.

Structure of the Document

This document is structured as follows:

- Chapter 1 introduces the research topic, its scope, the problem statement, and the key research questions.
- Chapter 2 provides background information on SD-WAN, QoS metrics, and the role of AI in modern network management.

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- Chapter 3 presents the literature review, including an analysis of academic work on QoS prediction in SD-WAN using AI and ML techniques.
 - Chapter 4 outlines the methodology, including dataset selection, feature engineering, model design, and evaluation metrics.
 - Chapter 5 presents the experimental setup and baseline model results on the Zenodo 5G dataset.
 - Chapter 6 details the project plan, timeline, tools, and supervision checkpoints.
 - Chapter 7 discusses the risks and limitations of the project, including data availability, model complexity, and real-world constraints.
 - Chapter 8 concludes the report by summarizing findings, highlighting real-world impact, and identifying directions for future work.

Chapter 2

Background

This chapter provides foundational knowledge needed to understand the scope and significance of this thesis. It covers the key concepts of Software-Defined Wide Area Networks (SD-WAN), Quality of Service (QoS) metrics in networking, and the role of Artificial Intelligence (AI) in network management. These concepts establish the context for the research discussed in later chapters.

2.1 Software-Defined Wide Area Networks (SD-WAN)

SD-WAN is an approach to wide area networking that uses software-based control to manage network traffic over multiple connection types, including MPLS, broadband, LTE, and 5G. Unlike traditional WANs that rely on expensive private circuits, SD-WAN provides centralized orchestration and greater flexibility, enabling cost-effective routing based on application priorities and real-time conditions.

A typical SD-WAN architecture includes:

- **Edge devices:** Deployed at branch locations to route traffic intelligently.
- **Central controller:** Makes policy-based routing decisions and pushes configurations to edge devices.
- **Overlay and underlay networks:** Logical tunnels (overlay) established on top of physical connections (underlay).

SD-WAN improves application performance by enabling dynamic path selection, fast failover, traffic shaping, and visibility across all network segments.

2.2 Quality of Service (QoS)

QoS in networking refers to the ability to provide different priority levels to different applications, users, or data flows, and to guarantee a certain level of performance for critical traffic. It is crucial for maintaining service-level agreements (SLAs), especially in modern, cloud-centric environments.

Common QoS metrics include:

- **Latency:** Time taken for a packet to travel from source to destination.
- **Jitter:** Variation in packet delay across a network.
- **Packet loss:** Percentage of packets lost during transmission.
- **Bandwidth:** Maximum data transfer rate of a link.

Effective QoS management ensures that delay-sensitive applications like VoIP and video conferencing are not negatively affected by congestion or fluctuating traffic.

2.3 Artificial Intelligence in Network Management

Artificial Intelligence (AI), especially Machine Learning (ML), is increasingly being used in networking to automate and optimize operations. AI techniques can analyze large volumes of network telemetry data to detect anomalies, classify traffic, and predict future network states.

In the context of SD-WAN, AI is being used to:

- Predict traffic trends and adjust policies proactively.
- Recommend optimal routing paths based on current and expected conditions.
- Forecast QoS metrics and warn of potential SLA violations.

AI models used for QoS prediction typically include supervised learning algorithms (like decision trees and neural networks) and reinforcement learning (used in adaptive routing and load balancing). These models require training on historical network data to make accurate forecasts.

2.4 Importance of QoS Prediction in SD-WAN

Modern enterprises rely on applications that require consistent and reliable network performance. Static or reactive QoS mechanisms are often too slow to prevent performance issues. QoS prediction enables a more proactive approach—by forecasting issues before they occur, it becomes possible to take preventive actions like re-routing traffic or scaling resources.

2.4 Importance of QoS Prediction in SD-WAN

This capability is especially important in SD-WAN environments where network conditions vary across different links and access types. Accurate QoS prediction can help optimize resource usage, reduce costs, and improve user experience.

Chapter 3

Literature Review

This chapter presents a detailed review of the existing research related to the use of Artificial Intelligence (AI) and Machine Learning (ML) techniques for predicting Quality of Service (QoS) in Software-Defined Wide Area Networks (SD-WAN). The aim is to understand what work has already been done, what methods and models have been proposed, and where gaps still exist. This will help assess the current state of the field and how it supports or challenges the research questions introduced earlier.

3.1 Search Strategy and Selection Process

To gather the literature for this review, a structured search was carried out using academic databases such as IEEE Xplore, Scopus, and Google Scholar. The following search terms were used:

- “QoS prediction” AND “SD-WAN”
- “AI for QoS” AND “Software-defined networking”
- “Machine learning traffic classification” AND “network performance”

3.2 Overview of Selected Papers

The search focused on papers published between 2018 and 2025 to ensure up-to-date relevance. Articles before 2018 were excluded unless they provided foundational knowledge. From an initial pool of documents, key research papers were selected based on the following criteria:

- Peer-reviewed publications
- Clear focus on QoS, SD-WAN, or software-defined networking
- Use of AI or ML for prediction, classification, or optimization
- Availability of experimental results, models, or architecture designs

Although the topic of QoS prediction using AI in SD-WAN is promising, the literature review revealed a limited number of high-quality research papers that directly focus on this specific intersection. Most existing studies either concentrate on general SDN environments or address QoS management without the use of AI or ML techniques. Nonetheless, this review provides valuable insights and highlights key research gaps that warrant further investigation.

3.2 Overview of Selected Papers

The selected papers cover a range of methods and applications—some focus on predicting traffic load, others on routing or load balancing. Table 3.1 categorizes these papers by their core approach and relevance to the research questions.

3.3 AI and ML Methods for QoS Prediction (RQ1)

Many studies have applied AI techniques to network traffic prediction and management. For instance, Rahman et al. (2019) [6] used Random Forest and deep

3.4 Suitability and Real-World Limitations (RQ2)

learning models to predict how many virtual network functions (VNFs) are needed during a traffic surge. Their models achieved over 95% accuracy and could prevent service degradation by scaling resources in advance.

Similarly, Zheng et al. (2022) [7] developed a deep neural network to classify traffic flows based on their latency sensitivity. They then routed this traffic using a QoS-Aware Routing algorithm that selected paths depending on the required delay tolerance. This combination of classification and path assignment showed how AI can directly improve QoS outcomes.

These studies clearly support RQ1 by showing that supervised learning and deep learning can be applied to anticipate traffic needs and assign optimal resources in advance.

3.4 Suitability and Real-World Limitations (RQ2)

Although the above techniques work well in controlled settings, there are challenges when using them in real networks. For example, the deep reinforcement learning (DRL) model by Dinh et al. (2024) [2] well in simulations and maintains safe traffic routing using control barrier functions (CBFs). However, DRL requires a lot of training time and GPU support, which may not be available in all SD-WAN deployments.

In contrast, Quang et al. (2023) [4] proposed SABE which is a lightweight method that uses queuing theory to estimate how much bandwidth is left on a link without sending test traffic. This approach avoids complex AI models but may be less flexible when traffic patterns become unpredictable.

This contrast shows that while AI models are powerful, their complexity and resource requirements must be carefully considered before deployment, which directly addresses RQ2.

3.5 Integration into SD-WAN Systems (RQ3)

Several papers also discussed how these models can be integrated into actual SD-WAN systems. For instance, Rahman et al. (2019) [6] simulated an SD-WAN scenario where their ML models helped reduce both QoS violations and cloud leasing costs by scaling VNFs only when necessary.

Quang et al. (2022) [3] presented a practical example of an enterprise moving from traditional MPLS to SD-WAN. Their central controller dynamically adjusted both routing and QoS scheduling based on the current network status. This shows that AI-based prediction and decision-making can be realistically applied in SD-WAN setups to improve efficiency and reliability.

These findings support RQ3, confirming that AI is not only a research tool but can also be integrated into live systems if the environment supports it.

3.6 Research Gaps and Observations

Despite the progress seen in these papers, several gaps were noticed:

- Most research focuses on predicting a single QoS metric like latency or bandwidth, rather than multiple metrics at once.
- Most results are based on simulations. Real-world deployment results in live SD-WAN systems are rare.
- Only few research documents are focusing on using Artificial Intelligence in the prediction or correction of QoS.
- Public datasets related to SD-WAN traffic are hard to find, limiting reproducibility.

- Few works compare the AI predictions directly against decisions made by standard SD-WAN controllers.

Filling these gaps could make QoS prediction models more robust, more generalizable, and more useful for industry.

3.7 Summary Of Literature Review So Far

This review found that AI and ML models especially supervised learning, deep learning, and reinforcement learning, are promising tools for predicting QoS trends in SD-WAN. The models help forecast traffic load, classify application needs, and adjust routing or resource allocation dynamically. However, practical deployment still faces challenges due to training complexity, hardware requirements, and lack of real-world validation.

Although the number of direct studies focused on AI-based QoS prediction in SD-WAN remains limited, the relevance of the topic continues to grow as networks become more software-defined and application-aware. Recent work such as MTEAL [1] also shows that state-of-the-art SD-WAN traffic engineering is moving beyond simple machine learning models towards graph neural networks, reinforcement learning, and multi-dimensional QoS optimisation. Based on the review and in consultation with Dr. Jawad Manzoor, this project will keep simple models as baselines only, while exploring stronger non-linear and deep-learning-inspired approaches where feasible. The scope will be carefully managed so that the chosen datasets, techniques, and evaluation metrics remain realistic within the resources and time frame of a master's-level project.

3.7 Summary Of Literature Review So Far

Table 3.1: Summary and Categorisation of Selected Literature

Paper	AI/ML Technique	Tech-	Main Focus	Related RQ(s)
Rahman et al. [6]	Random Forest, DNN		Predicting VNF requirements to maintain QoS and reduce cost	RQ1, RQ2, RQ3
Zheng et al. [7]	Deep Neural Network + ILP		Classifying application types and assigning QoS-based routes	RQ1, RQ2
Dinh et al. [2]	DRL (PPO, DDPG) with CBF		Safe traffic routing using AI that respects network constraints	RQ1, RQ2
Quang et al. [4]	SABE (Queuing theory-based estimation)		Estimating available bandwidth using passive measurements	RQ2, RQ3
Quang et al. [3]	Centralised Optimization + QoS scheduler		Managing QoS priorities and routing using SD-WAN controller	RQ2, RQ3
Chen et al. [1]	GNN + MARL		SD-WAN traffic engineering with multi-dimensional QoS metrics	RQ1, RQ2, RQ3

Chapter 4

Proposed Methodology

Based on the findings from the literature review and supervisor feedback, this project will focus on QoS prediction for SD-WAN-like network environments using public network-performance datasets. The primary goal is to predict QoS outcomes such as achieved throughput, latency, jitter, and packet loss from traffic load and network configuration features. The first implemented experiment uses the Zenodo 5G campus network dataset to predict achieved throughput, with bandwidth recommendation treated as a derived secondary target rather than the only objective.

4.1 Data Collection

This project will use publicly available datasets because a real enterprise SD-WAN dataset could not be obtained. The main public dataset is the Zenodo 5G campus network dataset, which contains throughput, jitter, packet loss, offered load, radio configuration, uplink/downlink direction, and packet-level one-way-delay measurements. Additional datasets suggested for exploration include the BNN-UPC Network Modeling Datasets and MAWI traffic traces. CICIDS2017

will be retained only as an exploratory proxy dataset because it was designed for intrusion detection rather than QoS prediction.

4.2 Feature Selection and Pre-processing

Common network parameters such as latency, jitter, packet loss, throughput, offered load, packet size, duration, flow count, protocol type, and radio/network configuration will be extracted where available. Data will be cleaned, normalized where appropriate, and split into training and testing sets.

The performance of any machine learning model in network traffic analysis depends heavily on the quality and relevance of the features extracted. I also plan to explore the below network features and parameters to explore and improvise the performance of the selected model. However, this heavily depends on the dataset availability.

Below Table explains the of shortlisted network parameters which often impact the performance in real world:

4.2 Feature Selection and Pre-processing

Table 4.1: Potential Features to be used for ML Model Training

Feature Name	Description
Latency	One-way packet delay derived from inter-arrival time (IAT) measurements
Packet Size (Average)	Mean size of packets in a flow
Flow Duration	Total time between the first and last packet in a flow
Packet Count	Number of packets in the flow
Bytes per Second	Average throughput in bytes
Packets per Second	Flow rate measured in packets
Protocol Type	Transport-layer protocol (TCP, UDP, ICMP)
Source/Destination Port	Application hint (e.g., HTTP = 80)
TCP Flags (SYN, ACK, FIN)	Used to identify connection status or handshake types
Inter-arrival Time	Time between packets in a flow
Flow Direction	Inbound or outbound traffic indicator
Window Size	TCP receive window size, indicating congestion levels
Flow Start Time	Time of day when the flow was initiated
Entropy of Packet Size	Statistical variability in packet sizes within a flow

These features will be preprocessed and normalized as needed before being used to train the selected machine learning models. Feature selection techniques such as correlation filtering and Recursive Feature Elimination (RFE) may also be applied to reduce redundancy and improve model generalization.

4.3 Machine Learning Models Selection

The primary objective of this project is to predict QoS outcomes from network and traffic features. In the Zenodo experiments, the first supervised regression target is actual achieved throughput in Mbps. A derived bandwidth recommendation percentage may also be evaluated as a secondary target, but care will be taken to avoid target leakage by removing actual throughput from the input features whenever that derived target is used.

To ensure robustness and compare model performance, multiple machine learning algorithms will be implemented and evaluated:

- **Linear Regression:** A simple and interpretable baseline model that assumes a linear relationship between input features and the selected QoS target.
- **Random Forest Regressor:** An ensemble model that handles non-linear interactions and feature importance effectively. It is well-suited for tabular data and robust against overfitting.
- **Gradient Boosting (e.g., XGBoost):** A powerful tree-based ensemble model that iteratively reduces prediction errors. It generally outperforms traditional models in structured data tasks.
- **Support Vector Regression (SVR):** A kernel-based method that performs well in lower-dimensional feature spaces. It will be tested as a supplementary technique.
- **Multivariate Regression:** If multiple QoS metrics need to be predicted simultaneously, multivariate regression will be used to capture inter-dependencies between outputs.

- **Feedforward Neural Network:** A shallow fully connected neural network will be considered to test whether non-linear feature interactions improve QoS prediction over traditional tabular models.
- **Advanced deep learning direction:** If time and data structure permit, transformer-inspired temporal models or graph-based approaches will be investigated conceptually or experimentally, motivated by recent SD-WAN and QoS optimisation work using GNNs and reinforcement learning [1].

Each model will be trained and evaluated using a consistent set of input features extracted from network flow data (see Table 4.1) and validated using performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score. Models will be selected based on both predictive performance and computational feasibility within the scope of the project.

4.4 Evaluation Metrics

The following metrics will be used to evaluate and compare the performance of different machine learning models in this project:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual QoS target values. It is easy to interpret and less sensitive to outliers.
- **Root Mean Squared Error (RMSE):** Measures the square root of the average of squared errors. This metric penalizes larger errors more heavily and is useful when large deviations are especially undesirable.
- **R-squared (R^2 Score):** Indicates the proportion of the variance in the dependent variable that is predictable from the input features. Higher values indicate better model performance.

- **Inference Time:** Measures the time taken by a trained model to make a prediction on new input data. This is critical for SD-WAN systems where real-time or near-real-time decisions are required.

Each model (e.g., Linear Regression, Random Forest, XGBoost, SVR, Neural Network) will be evaluated using these metrics to ensure a balanced trade-off between accuracy, interpretability, and speed.

Chapter 5

Experiments and Results

This chapter describes the experimental setup used for the first modelling stage and presents the baseline results obtained on the Zenodo 5G campus network dataset. The goal of this stage is to establish a reproducible benchmark that subsequent models can be compared against.

5.1 Experimental Setup

5.1.1 Dataset

The primary dataset used is the Zenodo 5G campus network dataset (record 13754300) [5], collected at two testbeds: the Norwegian University of Science and Technology (NTNU) and the University of Würzburg (WUE). The dataset contains iperf3 throughput measurements and one-way delay (OWD) packet captures under varied 5G radio configurations.

The throughput files for both testbeds were loaded and processed as follows:

- Each raw throughput row stores uplink and downlink metrics as paired columns. These were unpivoted into separate rows, one per direction, yield-

ing **604 rows** in total ($302 \text{ raw rows} \times 2 \text{ directions}$).

- OWD packet files (approximately 24 million rows combined) were aggregated per radio configuration to derive one mean latency estimate per configuration. These estimates were joined back onto the throughput rows.
- All feature columns were mapped to the project schema: `latency_ms`, `jitter_ms`, `packet_loss_percent`, `bandwidth_utilization_percent`, `actual_throughput_mbps`, `flow_duration_sec`, `packet_count`, `protocol`, `application_type`, `link_type`, `time_of_day`, plus Zenodo-specific configuration columns: `bw_mhz`, `slots`, `ratio`, `direction`, `offered_throughput_mbps`, `gnb`, `sdr`, and `testbed`.

The regression target for this experiment is **actual_throughput_mbps** (achieved throughput in Mbps). The derived column `recommended_bandwidth_percent` was excluded from model inputs to prevent target leakage, as it is computed directly from actual throughput divided by offered throughput.

5.1.2 Train/Test Split

The dataset was split into training and test sets using an 80/20 ratio with a fixed random seed (42) to ensure reproducibility. This produced **483 training rows** and **121 test rows**, with **17 input features** after dropping leakage and traceability columns.

5.1.3 Preprocessing Pipeline

A scikit-learn `ColumnTransformer` pipeline was applied before model training:

- **Numeric features:** Median imputation followed by standard scaling (`StandardScaler`).
- **Categorical features:** Most-frequent imputation followed by one-hot encoding (`OneHotEncoder`).

The pipeline was fitted on the training split only and applied consistently to the test split, preventing any data leakage from test rows into preprocessing.

5.1.4 Models

Two baseline models were trained:

- **DummyRegressor (mean strategy):** Predicts the training-set mean for every input. This serves as the lower-bound sanity-check baseline — any real model should exceed it.
- **Linear Regression:** A standard ordinary least squares model. Simple, interpretable, and fast. Used as the first real predictive baseline before non-linear models are introduced.

5.2 Results

Table 5.1 presents the evaluation metrics for both models on the held-out test set.

Table 5.1: Baseline Model Results on Zenodo 5G Dataset (Target: `actual_throughput_mbps`)

Model	MAE (Mbps)	RMSE (Mbps)	R^2	Inference (ms/row)
DummyRegressor (mean)	36.90	43.88	-0.007	0.015
Linear Regression	10.55	14.15	0.895	0.032

5.3 Analysis and Discussion

5.3.1 Linear Regression vs. Naive Baseline

Linear Regression dramatically outperforms the DummyRegressor across all metrics. An R^2 of 0.895 means the model explains approximately 89.5% of the variance in achieved throughput from the input features. The MAE of 10.55 Mbps indicates that, on average, predictions are within approximately 10–11 Mbps of the actual measured throughput.

In contrast, the DummyRegressor achieves an R^2 of -0.007 , confirming that predicting the training mean is no better than random chance for this dataset and provides a meaningful lower bound.

5.3.2 Inference Speed

Linear Regression requires 0.032 milliseconds per prediction, which is well within the latency budget of near-real-time SD-WAN QoS systems. This confirms that even at this baseline stage, the model is operationally suitable for deployment in reactive or predictive network control loops.

5.3.3 Feature Notes

Several feature columns in the Zenodo dataset take a constant value across all rows: `protocol` is always `UDP`, `application_type` is always `throughput_test`, and `link_type` is always `5G`. These columns contribute zero variance to the model and will be investigated for removal in subsequent experiments. The most informative inputs are expected to be `offered_throughput_mbps`, `bw_mhz`, `slots`, `ratio`, and `direction`, based on the domain knowledge that radio configuration and offered load directly constrain achievable throughput.

5.3.4 Dataset Size Limitation

The dataset contains only 604 rows after processing, which is small for ensemble or deep learning models. This constraint will be addressed in future work by incorporating OWD-enriched features and exploring additional datasets such as BNN-UPC and MAWI, as recommended by the project supervisor.

5.4 Planned Model Extensions

The baseline results presented above will serve as the comparison benchmark for the following models, which are planned for the next phase of experiments:

- **Random Forest Regressor** — an ensemble of decision trees that captures non-linear interactions and provides feature importance rankings.
- **Gradient Boosting / XGBoost** — an iterative boosting model that typically achieves strong performance on structured tabular data.
- **Support Vector Regression (SVR)** — a kernel-based method evaluated for comparison in lower-dimensional feature subspaces.
- **Feedforward Neural Network (MLP)** — a shallow fully connected network to assess whether non-linear activation functions improve prediction over the linear baseline.
- **Transformer or GNN-based approach** — motivated by state-of-the-art results in SD-WAN traffic engineering [1], these will be explored conceptually and experimentally if time and dataset size permit.

All planned models will be evaluated using the same train/test split, preprocessing pipeline, and metrics (MAE, RMSE, R^2 , and inference time) to ensure a fair and reproducible comparison.

Chapter 6

Project Plan and Timeline

This chapter outlines the structured plan and timeline for completing the capstone project. The schedule is aligned to the final submission deadline of 31st August 2026 and includes time for dataset validation, model development, evaluation, documentation, and supervisor feedback.

6.1 Project Milestones

The table below presents the proposed timeline of key project phases and deliverables:

Table 6.1: Timeline and Key Deliverables

Time Period	Status	Activities and Deliverables
June 2025	Done	Finalized project idea and submitted project proposal
July–Nov 2025	Done	Background research, literature review, and initial dataset exploration
Dec 2025–Jan 2026	Done	Finalized project scope, created GitHub repository, reviewed candidate datasets
Feb–Mar 2026	Done	Implemented preprocessing pipeline and baseline modelling scripts
Apr–May 2026	Done	Pivoted to Zenodo-first workflow; processed QoS features including OWD aggregation; ran Linear Regression baseline ($R^2 = 0.895$)
June 2026	Planned	Extend modelling to Random Forest, Gradient Boosting/XGBoost, SVR, and feed-forward neural network
July 2026	Planned	Analyse and compare all model results; explore BNN-UPC and MAWI datasets; write thesis draft; incorporate supervisor feedback
August 2026	Planned	Final proofreading, submit thesis by 31st August 2026, and prepare for viva

6.2 Tools and Resources

The following tools and platforms will be used throughout the project:

- **Programming:** Python (pandas, scikit-learn, XGBoost, TensorFlow/Keras)
- **Environment:** JupyterLab / VS Code
- **Data Analysis:** matplotlib, seaborn
- **Documentation:** Overleaf for LaTeX-based report writing
- **Version Control:** GitHub (for code management and backups)

6.3 Supervision and Checkpoints

Regular meetings and email check-ins with the supervisor, Dr. Jawad Manzoor, will be planned around key milestones. Ad-hoc consultation via email or MS Teams will be arranged when feedback is needed between reviews.

Chapter 7

Risks and Limitations

This chapter outlines the potential risks and limitations associated with the proposed project, along with mitigation strategies. Identifying these factors in advance helps to set realistic expectations and provides contingency planning to ensure the project stays on track.

7.1 Data Availability

One of the primary risks is the limited availability of real-world SD-WAN traffic datasets, particularly those with labelled QoS metrics. While public datasets provide valuable traffic data, they may not directly reflect enterprise SD-WAN-specific behaviour or QoS classifications.

An internal workplace dataset could not be obtained, so this project will rely on public datasets. This limitation will be explicitly discussed in the thesis to avoid over-claiming real-world SD-WAN generalisability.

Mitigation: The project will prioritize public datasets with direct QoS measurements, especially the Zenodo 5G campus network dataset. CICIDS2017 will be used only as an exploratory proxy dataset, while BNN-UPC and MAWI will

be assessed for possible additional experiments.

7.2 Model Complexity and Training Resources

Advanced models like deep neural networks or ensemble methods (e.g., XGBoost) may require significant computational power, especially for hyperparameter tuning and training on large datasets.

Mitigation: The project will begin with simpler models (e.g., Linear Regression, Random Forest), and only progress to more complex architectures if time and resources allow. Cloud-based resources (e.g., Google Colab, Kaggle Kernels) may be leveraged for experimentation if needed.

7.3 Feature Engineering Challenges

Not all features listed in the methodology may be available or relevant across all datasets. Inconsistent data schemas, missing values, or incompatible units may limit the effectiveness of certain features.

Mitigation: A flexible feature selection pipeline will be used. Feature selection methods like correlation analysis and Recursive Feature Elimination (RFE) will be applied to retain the most informative subset. Feature engineering techniques will be adapted depending on the final dataset used.

7.4 Evaluation in Real-World Context

Since the proposed work is likely to use offline datasets, the final results may not directly translate to real-time SD-WAN deployment without further validation. Simulations might not capture dynamic, production-grade behaviors such as live failovers or multi-path routing decisions.

Mitigation: The project will focus on building and validating a working proof-of-concept model using offline data. The limitations of generalization will be clearly documented, and suggestions for real-time deployment and testing will be proposed as future work.

7.5 Time Constraints

Any delays in data access, tool setup, or model convergence could impact the final delivery timeline.

Mitigation: A flexible and modular project plan has been developed, with early-stage tasks front-loaded to reduce later risk. Regular checkpoints with the supervisor will help track progress and address blockers quickly.

Chapter 8

Conclusion

This thesis has investigated the use of machine learning models to predict Quality of Service (QoS) outcomes in SD-WAN-like network environments, with a focus on achieving throughput prediction from radio configuration and traffic load features.

The literature review identified a broad range of AI and ML techniques applied to QoS prediction and SD-WAN traffic engineering, ranging from classical supervised learning to graph neural networks and reinforcement learning. The review confirmed that while deep learning and transformer-based models represent the current state of the art, interpretable baseline models remain essential as comparison benchmarks and are computationally feasible for near-real-time deployment.

The primary dataset used for experimentation is the Zenodo 5G campus network dataset. After processing, the dataset comprises 604 rows with 17 input features including radio configuration parameters, offered throughput, one-way delay, jitter, and packet loss. Baseline experiments show that Linear Regression achieves an R^2 of 0.895 and an MAE of 10.55 Mbps on the held-out test set, significantly outperforming the naive mean-prediction baseline ($R^2 = -0.007$). Inference time of 0.032 milliseconds per prediction is well within the requirements

of near-real-time SD-WAN control.

Data sourcing remains a limitation. An enterprise SD-WAN dataset could not be obtained from the workplace. CICIDS2017, initially explored as a proxy dataset, was found to be unsuitable for QoS prediction because it was designed for intrusion detection and contains attack traffic rather than QoS-labelled measurements. The project therefore relies on the Zenodo dataset as its primary source, with BNN-UPC and MAWI identified as candidates for further exploration.

The next phase of work will extend the evaluation to Random Forest, Gradient Boosting, SVR, and feedforward neural networks, with transformer or graph-based approaches investigated if time permits. All models will be compared against the Linear Regression baseline using the same split, preprocessing pipeline, and metrics to ensure a reproducible evaluation.

8.1 Real-World Impact or Benefit

The core value of this project lies in enabling proactive Quality of Service (QoS) management in SD-WAN environments through AI-driven predictions. In current networks, QoS violations are often addressed reactively, after user experience is already affected. This project aims to change that by predicting issues in advance using real-time flow-level features.

Practical Benefits in Real Networks:

- **Proactive Network Management:** Helps operators act before congestion or SLA violations occur, reducing downtime and customer complaints.
- **Cost Optimization:** Enables intelligent bandwidth allocation and resource scaling, potentially reducing overprovisioning or cloud leasing costs.
- **Improved QoE (Quality of Experience):** Real-time predictions allow

better support for critical applications like video calls, VoIP, and live collaboration tools.

- **Scalability:** The approach is suitable for large enterprise SD-WANs that span multiple locations and require dynamic QoS adjustments.

8.2 Future Work

Based on the current findings, future research should explore the following directions:

- **Multi-Metric Prediction:** Extend from bandwidth-only prediction to include latency, jitter, and packet loss simultaneously for holistic QoS control.
- **Hybrid AI Models:** Explore combining supervised learning with reinforcement learning for dynamic decision-making based on predicted states.
- **Real-Time Integration:** Integrate the predictive model directly into SD-WAN controllers or orchestration platforms to trigger policy updates.
- **Cross-Domain Generalisation:** Test the model on different network topologies and providers to assess adaptability across domains.

By addressing these improvements, the project could evolve into a plug-in or microservice within SD-WAN platforms that provides predictive QoS as a built-in capability.

8.3 Final Remarks

AI and ML hold significant promise for improving QoS in SD-WAN by enabling proactive, predictive management rather than reactive troubleshooting. While

8.3 Final Remarks

the current body of research is still emerging and somewhat fragmented, there is a strong foundation to build upon. With more focused work and collaboration between academia and industry, AI-driven QoS prediction can become a core feature of future network systems.

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